# SigA: rPPG-based Authentication for Virtual Reality Head-mounted Display

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## ABSTRACT

Consumer-grade virtual reality head-mounted displays (VR-HMD) are becoming increasingly popular. Despite VR's convenience and booming applications, VR-based authentication schemes are underdeveloped. The recently proposed authentication methods (Electrooculogram based, Electrical Muscle Stimulation-based, and alike) require active user involvement, disturbing many scenarios like drone flight and telemedicine. This paper proposes an effective and efficient user authentication method in VR environments resilient to impersonation attacks using physiological signals — Photoplethysmogram (PPG) , namely SigA. SigA exploits the advantage that PPG is a physiological signal invisible to the naked eye. Using VR-HMDs to cover the eye area completely, SigA reduces the risk of signal leakage during PPG acquisition. We conducted a comprehensive analysis of SigA's feasibility on five publicly available datasets, nine different pre-trained models, three facial regions, various lengths of the video clips required for training, four different signal time intervals, and continuous authentication with different sliding window sizes. The results demonstrate that SigA achieves more than 95% of the average F1-score in a one-second signal to accommodate a complete cardiac cycle for most adults, implying its applicability in real-world scenarios. Furthermore, experiments have shown that SigA is resistant to zero-effort attacks, statistical attacks, impersonation attacks (with a detection accuracy of over 95%) and session hijacking attacks.

## CCS CONCEPTS

• **Applied computing** → *Bioinformatics*; • **Security and privacy** → **Biometrics**.

## KEYWORDS

Biometric Authentication, Virtual Reality, Photoplethysmography, Physiological Signal, Head-mounted Display

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## 1 INTRODUCTION

With the advent of the metaverse era, virtual reality head-mounted displays (VR-HMD) become increasingly popular. According to [24], the global virtual reality market size will grow to $227.34 billion in 2029, with HMD contributing the largest revenue share at over 60%. Recently, VR-HMD-based applications have been rapidly penetrating other fields, like shopping in a virtual store [40], conducting complementary treatments in a virtual environment [43], carrying out military training in VR [13], and many alike. In most VR applications, VR-HMD stores a large amount of sensitive personal information. For example, credit card details are stored in the apps' cache; medical records are kept in the software tool; personal photos and videos are stored in the social media app. It poses a risk to the privacy and security of the user if not properly protected. Therefore, a robust user authentication mechanism is urgently needed for VR-HMD to prevent unauthorized access to stored information and minimize the risk of data breaches, identity theft, and other forms of privacy violations.

Unfortunately, VR-HMD authentication should not directly adopt traditional authentication methods like PIN codes and drawing graphics because of their low security levels. Specifically, PIN codes can easily be compromised by shoulder-surfing attacks [1]. Similarly, user behavior-based authentication on VR-HMD is vulnerable to mimicry attacks [73]. Furthermore, VR-HMD poses a severe challenge to its wearers against observers because most HMDs completely block the wearer's view, making it difficult for the user to observe the presence of an attacker or potential threat. A human visual system (HSV) has been proposed to use eye gaze patterns feature as an individual's unique identifier in VR authentication [39], but it requires active user participation. Transparent authentication refers to an authentication process that is seamless, effortless without requiring conscious effort or attention from the user. During a transparent authentication process, the user is not aware of the authentication process and can access the system or perform their tasks without interruption. For instance, in VR-based drone

piloting, the pilot must concentrate on the mission; otherwise, authentication that requires user participation may disturb pilots completing the mission. Last but not least, the limited computing resources of HMD devices restrict the development and deployment of traditional authentication systems.

As an alternative to traditional approaches, biometric user authentication based on physiological signals has received much attention, for instance, consumer-grade devices using electroencephalogram (EEG)-based biometric [67], a commercial ECG authentication solution by Nymi [19], and an electrical muscle stimulation (EMS) device embedded in a VR-HMD for user authentication by ElectricAuth [11]. Physiological signals have great advantages over traditional biometrics for authentication applications, as they are invisible to the naked eye, discouraging observation attacks. In particular, as a physiological signal, photoplethysmogram (PPG) has vast research potential due to its easy acquisition by optical methods [70]. With the aid of a light sensor, the PPG signal is captured from vascular flow information under the skin. Personal vascular flow information reflects the individual's cardiac health status and hemodynamic modeling, resulting in a unique identifier for authentication. Remote photoplethysmogram (rPPG) offers a significantly more convenient method. Using video cameras to capture subtle changes in skin color and blood flow, rPPG eliminates the need for direct contact with the skin. Nevertheless, rPPG has been overlooked by VR vendors and researchers while building user authentication for VR-HMD systems due to its current technological maturity stage.

In this paper, we propose a novel rPPG-based VR-HMD authentication system, namely SigA. SigA requires a face-oriented camera to be mounted inside the VR-HMD device to collect the PPG signal from the wearer's facial skin. The signal acquisition process requires no conscious effort or input from the user to promote a seamless VR experience without detracting from user immersion. The captured PPG signal with the skin information is transformed using a Spatio-temporal map (STM) to tolerate the large amount of noise in the PPG signal captured from the face. SigA combines the wearer's rPPG signal with their facial information to perform effectively on user authentication tasks. Due to the limited memory capacity of VR devices on the market, we employed transfer learning to guarantee low memory usage. In addition, SigA works on various time intervals, as short as 1 second, allowing continuous authentication in real-world scenarios. This paper's contributions are three-fold:

- We propose a novel user authentication system based on PPG signals, namely SigA. We are the first to combine rPPG signals and the Spatio-temporal map (STM) of the VR-HMD wearer's face video for user authentication. The utilization of STM can effectively mitigate substantial levels of noise present in the rPPG signal.
- Considering the limited resources inherent to VR devices, the integration of transfer learning was employed as part of our methodology. We comprehensively tested and compared the performance of several different image pre-training models for authenticating VR-HMD wearers. We demonstrated the high efficiency of the MobileNetV2 model that only requires one second long video clips and 13.6M of memory size to achieve an F1-Score of 0.95.
- We conducted an extensive reliability evaluation of the proposed SigA system and security analysis against impersonation attacks. Our method achieves an average recognition success rate of 0.95 for impersonation attacks.

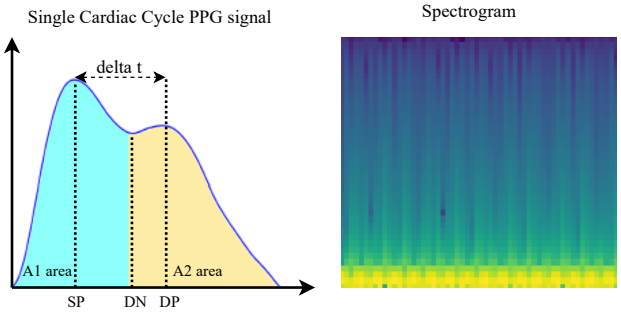



**Figure 1: Multiple signal features are used in PPG-based authentication. The left-hand side is the original PPG signal for a single cardiac cycle marked with the commonly used fiducial points, including SP: systolic peak index, DN: dicrotic notch index, DP: diastolic peak index. The right-hand side shows a spectrogram extracted from the raw PPG signal.**

The rest of this paper is organized as follows: The background of PPG-based authentication is given in Sec. 2. Sec. 3 describes our proposed PPG-based authentication scheme in VR-HMD. We provide a comprehensive evaluation of our proposed solution in Sec. 4. The security analysis of SigA is in Sec. 5, followed by a discussion in Sec. 6. We review related work in Sec. 7 and conclude the paper in Sec. 8.

## 2 BACKGROUND

### 2.1 Photoplethysmogram

Photoplethysmogram (PPG) measures the change in blood flow volume by the light reflected from the skin surface. Blood flow change reflects the subject's vascular dynamics and other unique physiological conditions [6]. Acquiring PPG signals is a relatively simple and inexpensive process that requires only a light source to provide illumination and a photodetector to detect minor differences in the amount of reflected light from the blood vessels [35]. This optical method is non-invasive and can be performed on various parts of the human body, such as the fingertip, face, or earlobe [9]. The pulsatile signal is the most prominent feature of the PPG waveform and represents the changes in blood volume caused by the cardiac cycle. It consists of a series of peaks and valleys [20]. The left-hand side of Fig. 1 shows the pulsating signal for one cycle. The highest peak represents systole (when the heart contracts and blood is pumped out of the heart), and the lowest valley represents diastole (when the heart relaxes and fills with blood). Because PPG signals can be acquired using simple and readily available equipment, they have become widely used in both medical and commercial applications.

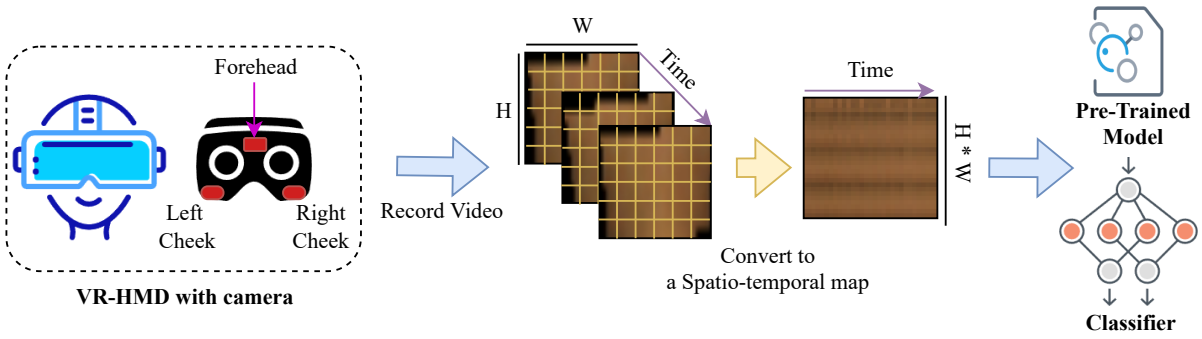



**Figure 2: Overview of our proposed VR-HMD authentication scheme (SigA). The VR-HMD captures video of the user's skin through an embedded face-oriented camera. The camera can be mounted on different locations of the VR-HMD, such as the forehead and cheeks. A stable light source is required to provide illumination. The video clip is then transformed into a Spatio-temporal map. The pre-trained model will extract the feature vectors of the Spatio-temporal map. Finally, the feature vectors are used in the training of the user authentication model. H: The height of the video frame is divided into $H$ equal parts. $W$ is the width of the video frame that is divided into $W$ equal parts.**

## 2.2 PPG-based Authentication

In earlier studies, the fiducial point in the original PPG signal was used as a unique identifier for user authentication [25]. As shown in Fig. 1, on the left-hand side, the systolic peak index and dicrotic notch index are associated with the individual cardiovascular status. The feature extracted from the PPG signal derivatives increased the distinguishability of the signal across subjects [32]. Spectrograms of PPG signals were used in [15] to obtain categorical features interpretable by a human observer. Impostors can be easily detected by comparing the differences between PPG signal spectrograms. Recent studies have used deep learning techniques to develop time-stable authentication systems [29].

## 2.3 Remote PPG

Remote PPG (rPPG) technology offers a new possibility to acquire PPG signals. Since the PPG signal is acquired by capturing the reflected light from the skin, it is present in the video clip. The PPG signal was extracted in [14] by analyzing the slight differences in skin color changes in continuous frames from the video. Hence, we only need a camera to capture the PPG signal. Earlier research extracted the PPG signal through the green channel in video clips because of its good tolerance to motion artifacts [63]. In contrast, the signal information is easily degraded in the red and blue channels. The periodic pulse signals can be separated from the different channels using Principal Component Analysis and Independent Component Analysis [5]. With the development of deep learning techniques, CNN-based models were implemented in [10] to recover PPG signals from face videos. Furthermore, Spatio-temporal maps were used to build deep learning networks to obtain a highly accurate acquisition of PPG signals from the subject's face [7]. However, directly applying rPPG signals to user authentication is challenging because of the impact of many environmental factors. Such as motion, lighting variations, ambient conditions, and camera quality can introduce distortions in the captured rPPG signals, compromising their reliability and accuracy [49].

# 3 THE METHODOLOGY OF SIGA

In this section, we propose a novel authentication scheme for VR-HMD using PPG signals, namely SigA. With a face-oriented camera, we capture a VR-HMD wearer's PPG signals and facial images to form a unique identifier for each user. SigA learns the identifier with respect to a specific user through a deep learning model. The trained classifier will serve as an authentication model in the end.

## 3.1 System Overview

In our authentication system, a face-oriented camera is mounted on the VR-HMD device close to the wearer's forehead or cheekbones so that the camera shoots the wearer's forehead and cheek regions. The forehead and cheek regions of a human face have a wealth of vascular distribution, yielding better rPPG signals [46]. Since each VR-HMD device almost always isolates external ambient light to provide its wearer with an immersive experience, our acquired PPG signals are mostly free of ambient light noise. We employ the STM representation to further mitigate the effects of noises other than ambient light. As shown in Fig. 2, VR-HMD captures a video stream of the region of interest (ROI) before converting it into an STM.

SigA follows the typical setup of a two-staged authentication process: 1) the registration phase and 2) the authentication phase. During the registration phase, a user simply wears the VR-HMD device for the camera to capture an STM of the user's skin. The STM is subsequently fed into SigA for training, allowing the system to learn the user's unique template. In the registration phase, the user keeps wearing the VR-HMD device to stabilize the camera. Since the camera stays relatively stationary to the wearer's head, the wearer can resume normal activities without remaining motionless. SigA will continue to train the users template in the background so that the templates are independent for each user. Similar to the registration phase, VR-HMD captures a temporal map of the wearer's skin and transmits it to SigA during the authentication phase. Essentially, authentication is regarded as a binary classification problem, which distinguishes whether a user is registered or

not. The whole authentication process is transparent to the wearer without requiring active participation. Hence, SigA is capable to support missions requiring full attention, like piloting drones. As the signals are captured continuously, our method is suitable for continuous authentication scenarios.

## 3.2 A Threat Model

We outline the threat model targeting the rPPG-based VR-HMD authentication model. The attacker attempts to bypass the authentication system on VR-HMD to access the wearer's private data. We assume that the attacker can physically access a victim's VR-HMD device and has sufficient time to compromise the device. If the VR-HMD device is shared among users, some curious users may access the victim's private information stored in VR-HMD without authorization. The malicious attackers pose a more significant security threat to VR-HMD due to their knowledge of the biometrics used for authentication and their ability to access the device legitimately. We further assume that the VR-HMD device is free of malware that monitors the wearer's input. To ensure the security of sensitive information during the authentication process, it is essential to prevent data from being captured by other suspicious processes. But the attacker can feed digital biometrics (e.g., images, video, signal sequences, etc.) directly to the authentication model.

## 3.3 Spatio-Temporal Map Acquisition

A spatio-temporal map (STM) represents the variations in the spatial and temporal characteristics of biometrics. It provides a wealth of dynamic information. The x-axis of the STM map represents the spatial features, while the y-axis represents the temporal characteristics, such as changes in the feature over time. It is often used in computer vision and video analysis to provide a compact representation [3]. In our paper, the STM represents a continuous signal through an image of $n \times m$ pixels to highlight the heartbeat pulse signal while suppressing other information unrelated to the pulse [48]. SigA uses STM as the unique identifier for a VR-HMD wearer. A face-facing camera mounted on the VR-HMD device captures the ROI region video. The light source can be provided by the display screen in the VR-HMD or by an additional set of LEDs.

The frame at the $t$th time is denoted as $F_t(C_r, C_g, C_b)$ with respect to the RGB channels. As shown in Fig. 2, each frame is split into $n$ equal regions, denoted by $\{R_{ft1}, R_{ft2}, R_{ft3}, \ldots, R_{ftn}\}$. $H$ and $W$ indicate that the height and width of the frame are divided into $H$ and $W$ equal parts, respectively. Hence, $n = H \times W$.

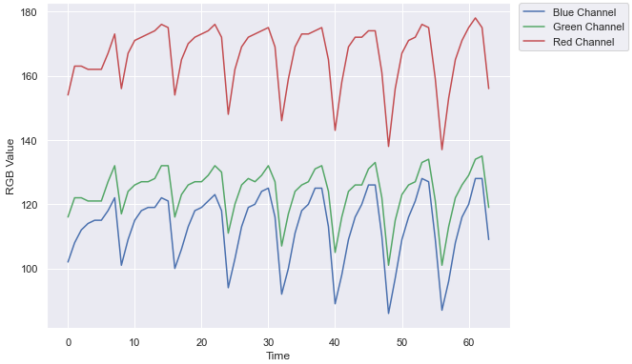
The $t$th column of the finalized STM consists of a series of arithmetic mean values of the split regions, denoted by $\{\bar{R}_{ft1}, \bar{R}_{ft2}, \bar{R}_{ft3}, \cdots, \bar{R}_{ftn}\}$. STM contains no information about facial components (the shape of the nose, lips, and eyes) except the pixel information of the skin. In practice, we use multiple skin-reflected light colors by changing the light source's color to enrich the uniqueness of the individual STM. Algorithm 1 lists the process to obtain the STM. Fig. 3 shows the STM's information about the PPG signal.

## 3.4 SigA: Authentication Model

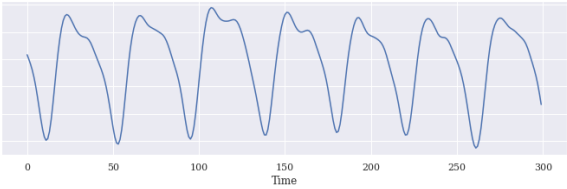
After obtaining the STM, we construct an image classification model to discern different users. It is a well-known fact that deep learning models require a substantial amount of training data to perform optimally, blowing up the time and effort for data acquisition. To overcome this challenge, we resort to transfer learning because knowledge learned during training on one task can be transferred and used to improve the performance on a similar but different task [47]. By using a pre-trained model as a starting point, transfer learning enables the model to learn faster and achieve better performance than starting from scratch with randomly initialized weights. We apply transfer learning to gain knowledge from multiple data domains [74], reducing the reliance on a large amount of target domain data. Hence, a pre-trained model is used in SigA to extract the features' representation. Image pre-training models are typically trained unsupervised in an extensive image database like

**Algorithm 1** Obtain a Spatio-Temporal Map (STM)

```
STM = []
% Read each frame from the video clip
for frame_t in video do
    % Store the STM for each frame
    STM_frame = []
    % Read sub-regions in the frame
    for sub_region in frame_t do
        STM_frame.append(sub_region)
    end for
    % Store STM for each frame
    STM.append(STM_frame)
    % Transpose to obtain the final STM
    STM_final = transpose(STM)
end for
```




**(a) Example of RGB channel values of STM with time axis.**




**(b) Reference PPG signals.**

**Figure 3: Comparison of the pixel value changes in the STM over time with the reference PPG signal.**

ImageNet[1], allowing the model to learn a general representation of an image. Specifically, we adopt a lightweight deep learning network MobileNetV2 [52] as the pre-training model with a trade-off between performance and model size.

Moreover, mobile devices are sensitive to computational complexity. We use depthwise separable convolution to reduce the computational complexity of the pre-trained model. According to [8], image transformation increases the invariance of the data learned by the deep learning model. Most pre-trained models' default input image size is $224 \times 224$ pixels. Therefore, we apply a random crop to process the image and resized it to $224 \times 224$ pixels before feeding the STM into the pre-trained model. Cropping randomly refers to the process of taking a portion of an image or a video by randomly selecting a region of interest and discarding the rest of the data. By cropping randomly in the image, our model learns STM at different scales, allowing it to obtain a good result in other sample sizes, even if it is trained in one-second samples.

After extracting the feature vectors of the STM by the pre-trained model, we add a fully connected layer as the classifier. Nadam [17] is used as an optimizer to accelerate the model converging. The distribution between positive and negative samples is unbalanced in our training data. We set the negative sample, non-registered users, as the side with the higher sample size. Negative samples are usually built into the initialized device. Positive samples from the registered users are usually collected during the registration phase of the authentication model. We balance the training set by sampling positive samples more than negative ones at the cost of increasing the registration time overhead. We also use focal loss [37] to mitigate the imbalance issue by changing the weights of positive and negative samples. Focal loss is calculated as follows:

$$Focal_loss = -\alpha(1-p)^{\gamma} log(p),$$

where $\gamma$ controls the shape of the weighting factor and $\alpha$ controls the focus of the weighting factor. That said, a larger $\gamma$ value results in a more steeply-sloped weighting factor, yielding more importance to hard examples (i.e., samples that are difficult to classify correctly). A larger $\alpha$ value assigns more importance to positive samples. By adjusting $\alpha$, the focal loss can be balanced between positive and negative samples. We finetune the parameters to optimize the performance using the method in [37].

After a result is generated for a single STM, SigA performs continuous authentication in a sliding window using majority voting rules. The formula for continuous authentication is shown as follows:

$$R_t = SigA(STM_t), \text{such that } R_t \in \{0, 1\},$$

where the sliding window length is $t$. The authentication model's result $R$ is calculated as follows:

$$R = \begin{cases} 0 & \text{if } MostCommon([R_1, R_2, \ldots, R_t]) = 0, \\ 1 & \text{otherwise,} \end{cases}$$

where $MostCommon(\cdot)$ is a function to find the maximum number of elements of the array. Whenever a sliding window outputs 0, the authentication fails. Because of this mechanism, SigA requires no active interaction between the VR-HMD wearer and the authentication system. SigA becomes user-transparent and potentially suitable for continuous authentication.

[1]https://www.image-net.org/

## 4 EXPERIMENT

This section reports a detailed evaluation of SigA. We used several rPPG datasets to simulate a VR-HMD environment. These datasets typically include a facial video clip with synchronized acquisition of reference physiological signals. We conducted multiple experiments on the UBFC-PHYS [45] dataset with various settings, including the use of different regions of the face (forehead, left cheek, right cheek), the length of the single sample video (1s, 2s, 5s, 10s), the different user states (Resting, Talking, Calculating), the length of the video required for training (10s, 20s, 30s, 40s, 50s, 60s, 120s), and nine pre-trained models (ResNet [26], MobileNetV2 [52], DenseNet [28], GoogleNet [60], InceptionV3 [61], VisionTransformer [16], ConvnextTiny [38], VGG [55], AlexNet [33]). We evaluated SigA on a few datasets (UBFC-RPPG [4], Cohface [27], PURE [59], LGI [50]). In addition, we assessed how SigA's strength against impersonation attacks. All experiments were performed at a sample video length of one second, except the single sample video length experiment. We used forehead as the region of interest, except for the comparison experiments with different facial regions.

**Table 1: Overview of the datasets used in our experiments.**

| DataSet | Participants | Resolution | FPS | Video Length |
|---|---|---|---|---|
| UBFC-PHYS | 56 | $1024 \times 1024$ | 35 | 180s |
| UBFC-RPPG | 50 | $640 \times 480$ | 30 | 60s |
| Cohface | 40 | $640 \times 480$ | 20 | 60s |
| PURE | 10 | $640 \times 480$ | 30 | 60s |
| LGI | 6 | $640 \times 480$ | 25 | 60s |

### 4.1 Datasets

We prepared a few common rPPG datasets before evaluating the performance of SigA. Each dataset includes multiple video clips of individuals' faces. Most video clips were illuminated by artificial light sources to reduce the interference of natural light. In all video clips, participants remained stationary most of the time. Tab. 1 summarizes the datasets used in our experiments.

- **UBFC-PHYS**[2] has the largest number of participants (56), the highest video quality ($1024 \times 1024$ resolution and 35 FPS), and the longest video duration (180 s).
- **UBFC-RPPG**[3] includes two sub-datasets. The first sub-dataset consists of eight participants, where each participant was recorded while remaining motionless. The second sub-dataset included 42 participants, simulating scenarios of interaction with the computer.
- **Cohface**[4] has 40 participants, each recording a one-minute video of their face under different lighting conditions. In the first condition, the subject's face is evenly illuminated by the

[2]https://ieee-dataport.org/open-access/ubfc-phys-2
[3]https://sites.google.com/view/ybenezeth/ubfcrppg
[4]https://www.idiap.ch/en/dataset/cohface

artificial light. In the second condition, natural light provides the illumination.

- **PURE**[5] consists of recordings of 10 individuals performing 6 distinct and controlled head movements in front of a camera. This includes rotating at different angles, and moving at different speeds.
- **LGI**[6] contains a total of 25 participants. Each participant in the database has recorded four separate sessions (Resting, Talking, Rotation, and Gym).

UBFC-PHYS contributed most cases in our experiments because it includes facial videos captured when users sat still in front of the camera. Participants of UBFC-PHYS have three states, including Resting, Talking, and Calculating. In Resting, the user remains stationary and does not talk. During Talking, the user stays in conversation with an experimenter. In Calculating, the user is asked to perform a calculation task and read the numbers out loud. For Cohface, we chose the data from the artificial light illumination scenario as it simulates the steady light source provided in the VR-HMD. The video clips in the Cohface dataset are highly compressed. For LGI, we used data from the publicly available 6 participants. The data in Gym is excluded. In PURE, UBFC-RPPG, we evaluate all scenarios together for each user.

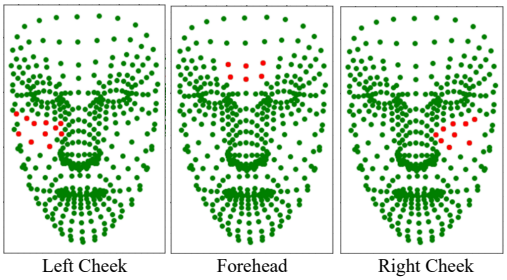



**Figure 4: Extracting different areas of the face. Green points correspond to 468 3D face marks. Red dots indicate the extracted areas.**

## 4.2 Experimental Settings

We implemented SigA in Pytorch[7]. The weights of the pre-trained model are directly derived from Pytorch's built-in functions. We modestly fine-tuned the pre-trained model's hyperparameters. Specifically, we set the model's epoch to 15 and all other parameters to default values. We used the Mediapipe[8] tool to carve human faces. These landmarks is typically used to represent a face in computer vision and facial analysis tasks [31]. It provide a rich representation of the facial features, such as the eyes, nose, mouth, and jawline. This allows us to separate the facial regions from other areas in each video clip. We selected the outer edge markers and discarded the inner markers with pyVHR[9] to remove the unwanted data outside the regions of interest. Finally, we extracted three regions per facial video clip: forehead, left cheek, and right cheek. As shown in Fig. 4, each extracted region is a contour surrounded by red dots. The STM will be extracted from the outer rectangle of these contour.

In our experiments, we divided our users into two categories: registered users (labeled as category 1) and non-registered users (labeled as category 0). Since non-registered user data far exceed users, we sampled one-twentieth of the non-registered user and 80% of the registered user data for training. If the model outputs a category with a probability greater than 0.5, that category is regarded as a prediction; otherwise, the category will be discarded. In the impersonation attack, we keep a portion of the non-registered users (10%, 20%, 30%, 40%, 50%, and 60%) out of the training to simulate attackers. For the statistical attack, we keep 30% of the non-registered users out of the training to comply with the best practice advocated in [23]. Accuracy is used to evaluate the performance against attacks and continuous authentication. We repeated the experiment for each user with the above settings. The final result is the average of all user results in the datasets.

**Table 2: Performance comparison of SigA on different Pre-trained models. '▼' indicates sorting from the largest to the smallest.**

| Pre-trained Model | F1-Score ▼ | Model Size (MB) |
|---|---|---|
| Resnet18 | **0.96** | 44.7 |
| **MobileNetV2** | 0.95 | **13.6** |
| Densenet | 0.95 | 30.8 |
| GoogleNet | 0.94 | 49.7 |
| InceptionV3 | 0.92 | 104 |
| VisionTransformer | 0.92 | 330 |
| ConvnextTiny | 0.88 | 109 |
| AlexNet | 0.82 | 233 |
| VGG16 | 0.79 | 528 |

## 4.3 Choices of Pre-trained Model

Tab. 2 compares the performance of the STM feature vectors extracted by different pre-trained models. Our original STM maps are very small (around 3KB with $36 \times 64$ pixels for one second of data) and have simple shapes (pixel extraction through the skin only). Most pre-trained models achieve an F1-Score of 0.9 or higher, indicating that they can extract useful features for the STM task. In general, better results (F1-Score greater than 0.94) for the feature vectors are extracted from the smaller model (model size less than 50MB). Large models introduce a huge feature space, making it difficult for the classifier to learn the subject's patterns. The F1-Score is only 0.79 for the largest model — VGG16 whose model size is 528MB. Resnet18 achieves the highest F1-Score as 0.96 with a model size of 44.7MB. MobileNetV2 achieves a similar F1-Score (0.95) for a model size of one-third of Resnet18. Based on these results, we choose MobileNetV2 as the pre-trained model in SigA, as it achieves a high F1-Score while having a relatively small model size, which may make it more suitable for real-world applications.

[5]https://www.tu-ilmenau.de/neurob/data-sets-code/pulse-rate-detection-dataset-pure
[6]https://github.com/partofthestars/LGI-PPGI-DB
[7]https://pytorch.org/
[8]https://google.github.io/mediapipe/
[9]https://github.com/phuselab/pyVHR

## 4.4 Choices of ROI

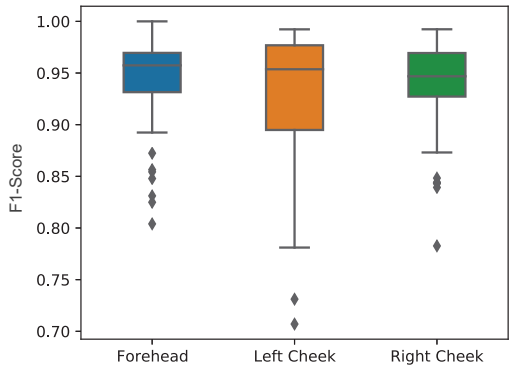



**Figure 5: Performance comparison of SigA on different facial regions.**

As mentioned in Sec. 3.1, the forehead, left cheek, and right cheek of the face are among the most common regions for extracting the signal of interest because they contain a rich distribution of blood vessels. To find the most suitable region for extracting STM, we compared the performance of the models trained in these three regions. As shown in Fig. 5, the average F1-Score is highest for the forehead region. The model's performance varies the most across users in the left cheek region. The highest F1-Score reaches 0.99, while the lowest is only 0.7. Moreover, in the forehead region, all users have a model performance higher than 0.8. Most of them are above 0.9. By analysis, we found that the forehead region's superior performance may be attributed to its relatively uniform skin tone. The shape of the extracted left and right cheek regions was more irregular than the forehead region. Also, the skin of the left and right cheeks is not in the same plane, making it more susceptible to the left and right head movements. Therefore, the STM collected in the forehead is the most suitable for our model among the three regions.

## 4.5 Time Efficiency

To explore the usability of our model, which is the time required for authentication, we conducted experiments with samples of different lengths of time (1s, 2s, 5s, 10s). We require at least one cardiac cycle to be included in STM. Since ordinary adults' resting heart rate ranges from 60 to 100 beats per minute [30], one second is the minimum time to finish sampling a cardiac cycle signal. Therefore, we set one second as the minimum time for user authentication. As shown in Fig. 6, the performance of the models between 1 and 5 seconds differs little (only 1%). However, the model performance of the F1-Score decreases by 10% in the sample length of 10s. The result contradicts with the intuition that longer sample lengths include more information that should better help us to distinguish between individuals. Through further analysis, we found that as the sample length increases, the morphological features of the PPG signal of a single heartbeat cycle become progressively less important for the authentication model. However, this morphological feature has an important contribution to the recognition of individuals. Although longer samples can potentially provide more information, they may also introduce a significant amount of additional noise, which can negatively impact the accuracy of the authentication model.

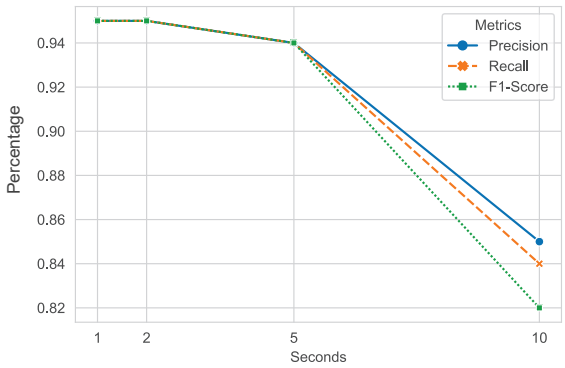
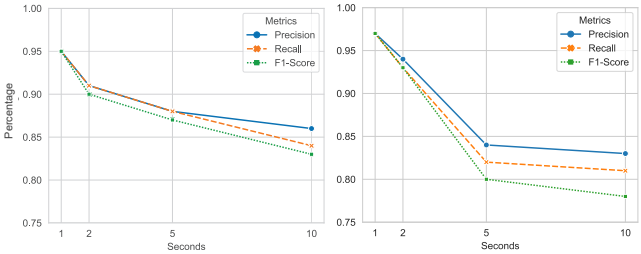



**Figure 6: Model performance in UBFC-PHYS with different sample lengths (1s, 2s, 5s, 10s). The model is trained separately on different sample lengths.**




**(a) Transforming images using Random Resized Crop.** **(b) Transforming images using Resize.**

**Figure 7: Model performance in UBFC-PHYS with different sample lengths (1s, 2s, 5s, 10s). The model is trained on a sample length of 1s.**

Fig. 7 shows the impact of Random Resized Crop on the model performance. Our models are trained on the images processed by Random Resized Crop and Resize method, respectively. For the Resize method, we use bilinear interpolation techniques to adjust the input size. Both models were trained on 1-second long video samples only. As shown in Fig. 7a, the F1-Score of the model with Random Resized Crop is reduced by 10% for the samples of length between 1s and 10s. However, the F1-Score of the Resize-processed model is reduced by 20%, indicating that our method has better adaptability in different sample time lengths. The Random Resized Crop method is more likely to obtain STM of similar scales in different time samples, providing better stability and consistency for model training.

In biometric authentication, the time required for model training is an essential factor to consider. Longer training times could improve the accuracy of the model. However, they also require more computing resources and may not be practical in real-world scenarios. To investigate the impact of training time on model performance, we conducted experiments with our authentication model using different training times, ranging from 10s to 120s.

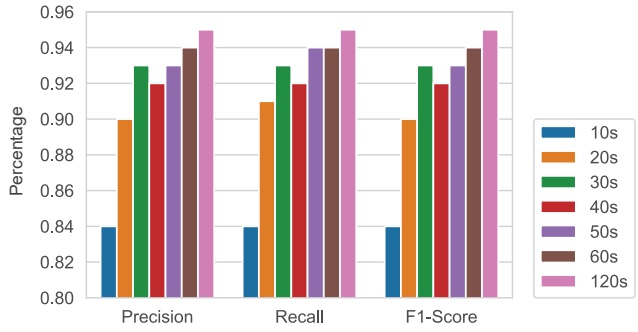



**Figure 8: Comparison of model performance for different training times (10s, 20s, 30s, 40s, 50s, 60s, 120s). The time duration for a single sample is 1s, indicating that the 10s training time contains 10 training samples.**

Fig. 8 shows that the model performance increases with training time. In particular, between 10s and 30s, the F1-Score of the models differs by almost 20%. Interestingly, the change in model performance is not significant after 60s, with the addition of a minute of training time increasing by just 1%. It indicates that while increasing the training time can lead to a significant improvement in the model's performance, there is a point of diminishing returns beyond which additional training time has little impact on the model's performance. Therefore, one minute of training time is sufficient to achieve high accuracy in our authentication model. We also found that the Recall of the model in identifying non-registered users did not vary much between the training times. Our model can achieve the desired result for non-registered user recognition with a short training time (less than 20s). The non-registered user recognition is a critical aspect of any biometric authentication system, as it prevents unauthorized users from gaining access. It indicates that our model can achieve a good trade-off between training time and model performance in one-minute training time.

## 4.6 Authentication Performance

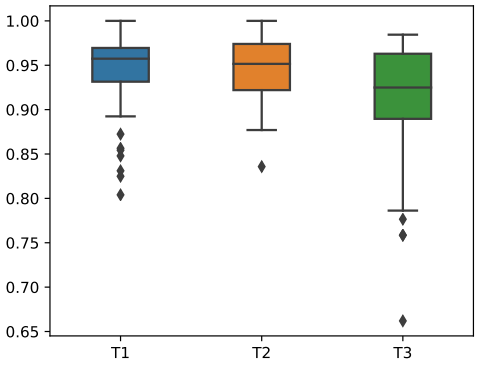



**Figure 9: Performance comparison of SigA on different user status. T1: Resting, T2: Talking, T3: Calculating.**

PPG signal is considered to be associated with autonomic nervous system activity [45]. Autonomic nervous system activity is a major component of the emotional response [58]. Changes in mood are reflected in the morphological characteristics of the PPG signal. We investigated the effect of STM collected under different user statuses (T1: Resting, T2: Talking, T3: Calculating) on the authentication model. From Fig. 9, we find that the average F1-Score of the model in both T1 and T2 is around 0.95. Since the STM was extracted from the forehead, the model did not capture lip movements during the conversation. Subtle changes in the head did not affect the model's performance. In T3, the calculating status impacts the model more significantly. The average F1-Score among users in the calculating status is 4% lower than that of users in the other two statuses. The lowest F1-Score of one user in the calculating status is 0.66. The results show that the calculating status subtly affects the model's performance.

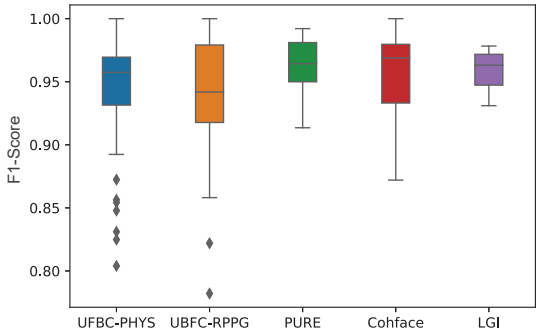



**Figure 10: Performance of SigA in different databases.**

**Table 3: Performance comparison with other VR authentication methods. Auth: Authentication.**

| Methods | FAR (%) | FRR (%) | F1-Score | Auth time (s) |
|---|---|---|---|---|
| OcuLock [39] | 3.550 | 3.550 | 0.983 | $\leq 10$ |
| SkullConduct [53] | 6.900 | 6.900 | - | $\leq 23$ |
| Brain Password [36] | 2.500 | 2.500 | 0.955 | $\approx 4.800$ |
| SoundLock [71] | 0.760 | 0.910 | 0.984 | $\leq 7$ |
| SigA (Ours) UBFC-PHYS | 3.774 | 3.774 | 0.955 | $\approx 1.200$ |
| SigA (Ours) UBFC-RPPG | 4.550 | 4.550 | 0.945 | $\approx 1.200$ |
| SigA (Ours) PURE | 2.863 | 2.863 | 0.960 | $\approx 1.200$ |
| SigA (Ours) Cohface | 1.810 | 1.810 | 0.965 | $\approx 1.200$ |
| SigA (Ours) LGI | 2.603 | 2.603 | 0.960 | $\approx 1.200$ |

To extensively assess the feasibility of SigA, we have evaluated it on several common datasets. Fig. 10 presents our experimental results on the five datasets (UBFC-PHYS, UBFC-RPPG, PURE, Cohface, and LGI). Our model achieves high performance for all datasets, with an average F1-Score above 0.945. Even for highly compressed data in the Cohface, SigA achieves a median F1-Score

**Figure 11: Examples of facial expression changes. The bottom row shows the image when the user is expressionless. The top row shows the image when the corresponding users have expressions or head moves.**

of 0.965. While the overall performance of SigA was high on the evaluated datasets, the F1-Scores for several users in the UBFC-PHYS and UBFC-RPPG datasets were below 0.88. To understand why some users had a lower F1-Score, we observed the authentication sessions in detail. It was found that frequent and significant head movements during the authentication period can affect the authentication model's performance. The model relies on detecting specific facial landmarks and features to identify the user, and head movements can cause these landmarks and features to shift, making it harder for the model to identify the user correctly. Moreover, facial expressions affect the authentication model's performance, such as frowning in anger and raising an eyebrow when surprised. These facial expressions can cause the skin to crease, affecting the reflection of skin light, which leads to a change in the PPG signal captured by the video sequence. To further illustrate this issue, Fig. 11 demonstrates the apparent variation in the reflected light on a user's face during an authentication session while the user's head was moving.

For comparison with the existing VR authentication methods, we present the Equal Error Rate (EER) of SigA from different datasets. EER provides an objective measure of the point where the false acceptance rate (FAR) and false rejection rate (FRR) are indistinguishable. Notably, as shown in Tab. 3, our lowest EER achieved a rate of 1.81%. Though SigA's EER is higher than SoundLock's EER, SoundLock requires a quiet environment for accurate authentication without being transparent to the user. In contrast, our proposed method, SigA, offers a definite advantage in terms of both transparency and authentication time.

**Table 4: Performance comparison between Original and ONNX Model. O: Original Model.**

| Class | Precision | Recall | F1-Score |
|---|---|---|---|
| O-User | 0.96 | 0.96 | 0.96 |
| ONNX-User | 0.96 | 0.97 | 0.96 |
| O-Non-User | 0.90 | 0.89 | 0.90 |
| ONNX-Non-User | 0.91 | 0.89 | 0.90 |

To make the model more portable, we exported the original model to Open Neural Network Exchange (ONNX)[10] format. It is an open file format for storing trained models. ONNX allows users to train and deploy machine learning models using a variety of frameworks and then convert those models to the ONNX format for seamless integration with other frameworks and platforms. Tab. 4 shows no significant difference in the ONNX model compared to the original model for identifying users and non-users, indicating that we can easily migrate this setup to VR-HMD devices.

[10]https://onnx.ai/

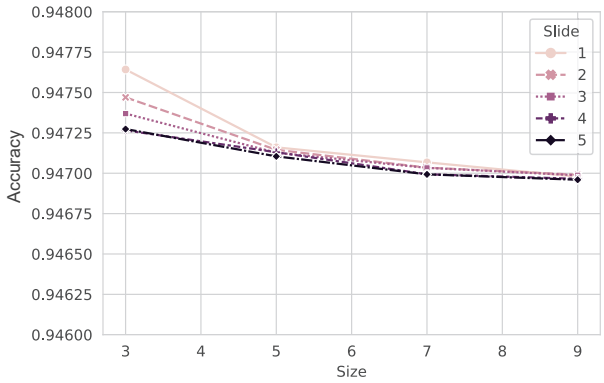



**Figure 12: Accuracy of the SigA model in continuous authentication. Size: Size of sliding window. Slide: Distance per movement of the sliding window.**

As described in Sec. 3.4, SigA can perform continuous authentication. The sliding window and slide size are essential parameters in continuous authentication, as they define the data sample size used for authentication and the frequency of updates to the data sample, respectively. To identify the most suitable values for these parameters, we conducted a series of experiments with different settings, varying the size and slide step of the sliding window. Fig. 12 shows the performance of SigA in continuous authentication. Notably, we observe that SigA's accuracy remains stable as the size and slide step of the sliding window vary. SigA's accuracy consistently remains above 0.946.

## 5 SECURITY ANALYSIS

This section presents a comprehensive security analysis of SigA. We have carefully examined potential attack threats that could compromise the system's security to achieve this objective. Additionally, we have evaluated the effectiveness of security in SigA to protect against these threats.

### 5.1 A Threat Analysis

Based on the threat model described in Sec. 3.2, the authentication model is vulnerable during the authentication and post-authentication phases. In the authentication phase, an attacker may attempt to bypass the authentication system by using fake inputs. For the post-authentication phase, attackers can exploit user sessions retained after user login to steal confidential information. In this paper, we will meticulously examine the following widely encountered and potentially devastating attacks:

- **Zero-Effort Attack**: Without prior knowledge of the enrolled biometrics, a cunning adversary can still attempt to

gain access to sensitive information or resources by randomly guessing. These attacks can be time-consuming and resource-intensive, but they can still pose a significant threat, especially if the biometric system has a low level of complexity. Furthermore, the authentication system should prevent accidental logins from unauthorized users.

- **Impersonation Attack**: An attacker can attempt to access the user's authentication model by mimicking the user's biometrics. In our scenario, an impersonation attack occurs when an attacker mimics the user's authentication process by using their facial information during authentication, especially for internal users who have some knowledge of our authentication system.
- **Statistical Attack**: An adversary has access to a substantial volume of biometric data and is aware of the features used by the authentication system, and can carry out a statistical attack. It involves performing a comprehensive statistical analysis on the dataset and deriving the probability distribution for each feature [54]. Armed with this information, the adversary can then create synthetic biometrics that follows these distributions in an attempt to bypass the authentication system and gain unauthorized access to sensitive information and resources.
- **Session Hijacking Attacks**: The attacker can steal the session ID or take over an already-established session between the user and the devices. In reality, users may leave their devices on their desks after logging in, and a malicious attacker could skip the initial login step and access sensitive information directly.

## 5.2 Zero-Effort Attack

The Zero-Effort Attack is the simplest, attempting to compromise an authentication system simply by random guessing. In SigA, the input to the model is an RGB image of size $t \times n$. Assume the model uses a 1-second STM. $t$ denotes the frame rate of the captured video. RGB takes values in the range of 0-255. The attacker's goal is to accurately guess the value of each pixel in the STM of a registered user. The probability of an accurate guess for an STM pixel is:

$$p_{tn} = 1/256^3$$

$t, n$ denote the position of the pixel in the STM. The probability of guessing the value of all pixels becomes:

$$p_{total} = 1/(256^3)^{t \times h \times w}$$

$h$ and $w$ represent the height and width of the ROI area, respectively. We assume $t$ to be 25, which is the smallest frame rate in the datasets we have used. A large $t$ value will make it more difficult to succeed in an attack. For $n$, we also assume that its value is 25. It consists of a $5 \times 5$ region, representing only one ten-thousandth of a video clip with a resolution of $640 \times 480$. RGB takes values in the range of [0,255]. The probability of an accurate guess for an STM pixel is $p_{tn} = 1/256^3$. If we half the range of possible values for the skin pixel color, the probability of guessing the whole STM is $p_{total} = 1/(128^3)^{25 \times 25}$. It is challenging to achieve a successful attack if it only requires a 50% accuracy rate. Though many attempts can be automated, we can protect the system by limiting the number of failed attempts for a real-world authentication system.

## 5.3 Impersonation Attack

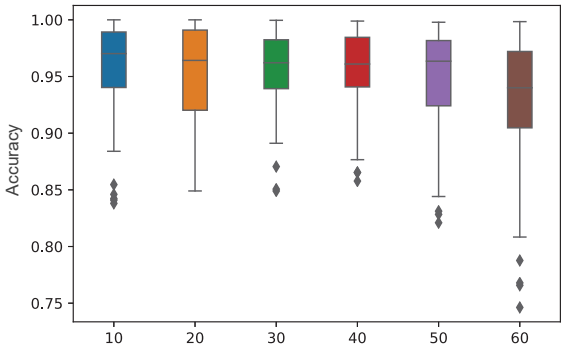



**Figure 13: Accuracy of the SigA model in recognising impersonation attacks. The $x$-axis indicates the percentage of potential attackers out of total users.**

Impersonation attacks are the commonly used attacks against VR-HMD authentication because of their simplicity [39]. In our authentication solution, the facial information captured by the camera is obscured by the VR-HMD device, which an external attacker cannot access in real-time. But they can still launch impersonation attacks. The attacker is not required to obtain the STM information of the VR-HMD wearer in advance. The attacker needs to steal and wear the VR-HMD before imitating a registered user to access the system. In our experiments, the other non-registered users were treated as attackers when a specific user was selected as the registered user. To prevent the model from gaining the knowledge of the attacker's choice, we randomly excluded 10%, 20%, 30%, 40%, 50%, and 60% of the users in the training set, respectively. The non-registered users were regarded as attackers. We repeated the experiment for all users in the UBFC-PHYS dataset. As shown in Fig. 13, SigA recognizes impersonation attacks perform consistently well, with a success rate of around 0.95.

## 5.4 Statistical Attack

In this attack, we assume that the attacker has access to the STM information of many non-registered users. Although it would be difficult for an attacker to obtain STM information from registered users, it may be possible to generate forged information by estimating the distribution of features of other users and then using the most likely feature values. As shown in Figs. 14 and 15, we counted each pixel value of the STM for all the users who are not involved in training and calculated their most likely values. We also calculate an STM for the attack by averaging over each pixel. In our results, the success rate for identifying an average STM attack is 95%; the success rate for identifying a statistical STM attack is 100%. Our results indicate that it is difficult for either the average STM or the statistical STM to compromise our authentication model. Since STM includes the user's continuous physiological signal, the template obtained by averaging will flatten the signal's waveform. On the other hand, the STM obtained by statistical methods will truncate in the continuous signal. It leads to the rejection of the

authentication model. Thus, estimating the STM for a specific user with the distribution of features of other users is unrealistic.

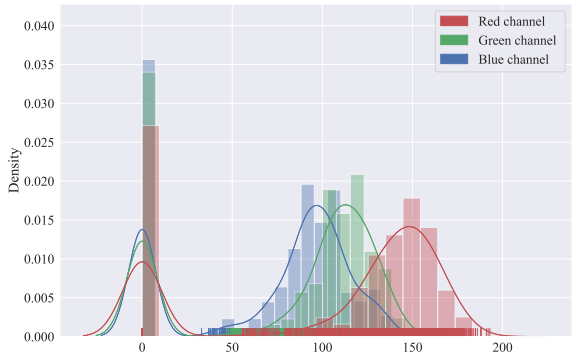



**Figure 14: Example of the distribution of different color values in the first pixel of STM.**

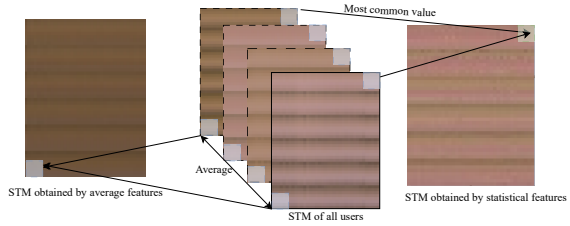



**Figure 15: Example of the distribution of different color values in the first pixel of STM.**

## 5.5 Session Hijacking Attacks

Session Hijacking Attacks typically occur in web scenarios when an attacker intercepts a user's session ID and uses it to impersonate the user, effectively taking over their authenticated session. This attack can also easily occur in a traditional user authentication scenario. In one-time authentication, a session hijacking attack can occur when a user leaves the device after logging in, and the user's credentials remain for a while. By employing our solutions for continuous authentication, it is possible to eliminate session hijacking attacks. Despite the attacker gaining access to the user's device and being unable to pass authentication within a specific time window, the system will prevent the illegal user from continuing access.

## 5.6 Entropy Analysis

Entropy measures a random variable's average amount of information or uncertainty. Since it quantifies the unpredictability or randomness of data [12], entropy is frequently employed in biometric systems to evaluate the security strength of authentication methods. By measuring entropy, the level of uncertainty or randomness in the biometric data can be quantified, providing insights into the information content and unpredictability of the authentication factor. Higher entropy values generally indicate stronger

**Table 5: Entropy of various authentication methods.**

| Authentication method | Entropy (bits) |
|---|---|
| Password [64] | 20 – 23 |
| PIN (4-digit, 6-digit) [65] | 8.41, 13.21 |
| Keystroke [51] | 3.48 – 4.62 |
| Iris [68] | 278 – 288 |
| Fingerprint [62] | 18.60 |
| Face [2] | 37.00 – 55.60 |
| **SigA (Ours)** | **21.50** |

authentication methods with greater resistance to unauthorized access or spoofing attempts. In this paper, we follow the suggestion in [71] and use relative entropy as a measure. Relative entropy, also known as Kullback-Leibler divergence (KLD), measures how one probability distribution $P(x)$ diverges from another reference probability distribution $Q(x)$. $P(x)$ represents the feature distribution of the enrolled user, and $Q(x)$ denote the distribution of other users. As shown in Tab. 5, we compare relative entropy among different authentication factors. SigA exhibits a higher relative entropy value than fingerprints and demonstrates comparability with the relative entropy of passwords. While face and iris authentication methods exhibit the highest relative entropy, installing traditional face and iris authentication systems on VR devices can be challenging due to the potential occlusion caused by the HMD and the iris scanners' cost constraints. Hence, SigA has great potential to become a reliable and secure authentication factor.

## 5.7 Privacy

In addition to the data stored by the user on the device, biometrics raises a significant threat to privacy. For example, the General Data Protection Regulation (GDPR)[11] is designed to protect the privacy and personal data of EU citizens. SigA considers a privacy concept named feature anonymity that is also known as irreversibility. Formally, an attacker should not be able to infer the features of a partial face through STM. Some work has demonstrated a mapping between the gradients updated by the model and the data involved in the training, i.e., some or even all of the information from the training data can be stolen [72]. We assume that the attacker can obtain the registered user's STMt at the moment $t$. The attacker then tries to obtain the real identity of the registered user (e.g., their face). In SigA, facial features from the areas like eyes, nose, and mouth with distinct shapes are excluded to prevent the attacker from reversing the user's identity. Additionally, SigA uses the mean value of the pixels in the sub-region for the value of each pixel in the STM, which further blurs the detail of the facial skin. Fig. 3 shows that the primary information stored in the STM comes from the PPG signal.

# 6 DISCUSSION

The computation time required for a practical authentication method is a critical factor. A system that requires excessive computation

[11] https://gdpr-info.eu/

time to verify a user's identity may frustrate users and discourage them from using the system. Traditional PIN codes usually require 2-3 seconds of authentication, which is considered quasi-Efficient-to-Use [57]. In this paper, we focus on the efficiency of the recognition phase of the authentication, as the registration phase usually only needs to be carried out once. In SigA, the time overhead usually consists of acquiring the original video, computing the STM, and model inference. In the acquisition video, as discussed in Sec. 4.5, 1 second of information is sufficient for our model to obtain accurate authentication results. When calculating the STM, since the VR-HMD does not require facial recognition, we exclude the processing time of Mediapipe. The final average processing time per frame is only 1.7ms. Even with a high frame rate camera, like 120Hz, it only takes 204ms. For model inference, We take an average of 8.3ms on the Tesla T4 GPU. Overall, the inference time of the model is between 1 and 1.2s, depending on the camera frame rate. Considering that our model is transparent to the user and does not block the user's operation, an authentication time of around 1 second is acceptable.

We discuss four common attack threats in Sec. 5. Presentation attacks are also one of the most common types of attacks in biometrics. An attacker attempts to impersonate a legitimate user by presenting false biometric data to an authentication system, such as fake fingerprints, masks, or voice recordings, to mimic the characteristics of genuine biometric information. An attack against a PPG-based user authentication system was proposed in [34] using the rPPG captured in the video. Nevertheless, video quality and motion limitations can significantly affect the reliability of spoofing attempts. Lower video quality and resolution can introduce noise, artifacts, and loss of details, making it more challenging for an attacker to accurately replicate the user's physiological characteristics. Videos may easily suffer from motion blur or artifacts, especially when the user or the camera is moving. These artifacts can distort the user's physiological signals, preventing an attacker from reproducing PPG signals accurately. In our scenario, the attacker needs to obtain the STM information of the user. First, since VR-HDM covers the recognition area, it is hard to obtain the STM of the user in real time. Note that we do not consider channel-based attacks since the channel is appropriately encrypted. Second, even if an attacker obtains a facial video of a victim, the rPPG technique's amplitude is comparatively small and can be influenced by various factors, such as external lighting, head movement, device interference, and others [44]. For the data in our experiments, participants were usually one meter away from the camera. Settings beyond one meter will significantly affect the acquisition of the rPPG signal [56]. In practice, the camera is usually closer to the recognition area, and it is challenging for the attacker to get more precise information about the target without being detected. Moreover, in the future, we can further change the light source to provide more recognition information, such as wavelength and intensity. When the light source color changes, it affects how the face reflects light, leading to changes in the distribution of the STM's red, green, and blue channel values.

Despite the encouraging contributions, there are still limitations to our work. Due to equipment limitations, we have not deployed our method on a real VR-HMD. Deploying SigA on a physical VR-HMD would be cost-intensive. The expenses include the potentially significant costs associated with the software development, optimization, and adaptation of our method for various VR platforms. SigA involves video capture of the user's face, implying that it requires integrating additional hardware beyond the ordinary VR-HMD. We must admit that integrating such additional hardware poses challenges, including engineering cost, technical feasibility, and practical considerations. Moreover, deploying experimental methods on physical devices, especially those involving user interactions or sensitive data, may raise ethical concerns. Ensuring user safety, privacy, and data protection will require additional measures and approval from ethical review boards or institutional protocols, which is out of this paper's scope. Considering that our research is currently in the initial stages, we prioritized demonstrating the viability and effectiveness of our proposed authentication method under controlled conditions. We performed simulations using the existing dataset. While the simulation may not have been a perfect substitute for real-world conditions, it still provided valuable insights into VR-HMD authentication. We will conduct a follow-up study on real devices in the future.

## 7 RELATED WORK

There are many authentication methods for VR-HMD. As one of the oldest types of authentication methods, VR-HMD users often enter passwords/PINs through a virtual interface via a VR controller to authenticate themselves [41]. Alternatively, VR-HMD users swipe a predefined pattern like many swipe authentication methods used on mobile phones [69]. Unfortunately, the entire authentication process is typically visible to the public, which could allow an adversary to observe. An attacker can infer the password/PINs from the victim's shoulder movements [22]. Even with SMS-based two-factor authentication, recent studies have revealed that these measures alone are insufficient to provide a comprehensive defense against the constantly evolving range of threats [66]. In our method, the VR-HMD fully covers the area around the user's eyes, and the attacker cannot observe directly from the external side. While rPPG signals can be captured remotely, it is highly sensitive to environmental factors such as lighting conditions and distance. As discussed in Sec. 6, this sensitivity makes it challenging for an attack without the victim's awareness. Moreover, 3D PINs [42] and gaze-based authentication [21] are proposed to mitigate the risks of shoulder surfing attacks. But the HMD's virtual keyboard is very cumbersome to use. The gaze-based approach requires the eye to be positioned on the target space and remain there for a certain amount of time to be considered correct input. They significantly prolong authentication time and reduce the availability of the solution [18]. As discussed in Sec. 6, our authentication time can be completed in around 1 second.

Physiological signals as biometric features have recently been used to authenticate VR-HMD users, like electroencephalogram, electromyogram, and electrooculogram [11, 36, 39]. Physiological signals have obvious advantages over traditional biometric authentication methods because unique and individualized biometric features are generated by complex visual or electrical stimuli. Despite these advantages, the existing physiological signal-based authentication methods demand the wearer's full attention, which disturbs some jobs like drone pilots. In contrast, our approach does not

require the user to perform specific actions. The rPPG signal measures the changes in skin blood flow. It can be captured as long as the user's circulatory system functions, allowing for non-intrusive and continuous measurement of their physiological state. Furthermore, our approach does not require additional hardware beyond a camera, making it convenient and cost-effective.

## 8 CONCLUSION

With the popularity of VR-HMD, traditional authentication methods are challenging to apply in emerging scenarios. As the potential for applying physiological signals in user authentication continues to be explored, we propose an effective and efficient VR-HMD authentication system based on PPG signals, namely SigA. SigA combines the subject's PPG signal and facial video using STM to ensure effectiveness and efficiency. In our experimental results, SigA only requires a training sample length of 60s and a single sample length of one second during the authentication. SigA performs consistently well on different datasets, including highly compressed data (Cohface). Though changes in facial expressions may affect the model's accuracy, SigA maintains an F1-Score of over 0.95. SigA requires a little storage room as its pre-trained model is only 13.6MB, thanks to MobileNetV2. Although the user's status affected the model's performance in the Calculating scenario for the UBFC-PHYS dataset, the overall performance was only reduced by 0.02 compared to the resting status. Furthermore, our model consistently achieves high accuracy rates in continuous authentication and can identify 95% of illegal accesses in the face of impersonation attacks. SigA can be also extended to support continuous authentication. In the future, we will employ cryptographic primitives to secure SigA from being monitored and try to deploy SigA on real devices for further validation.

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